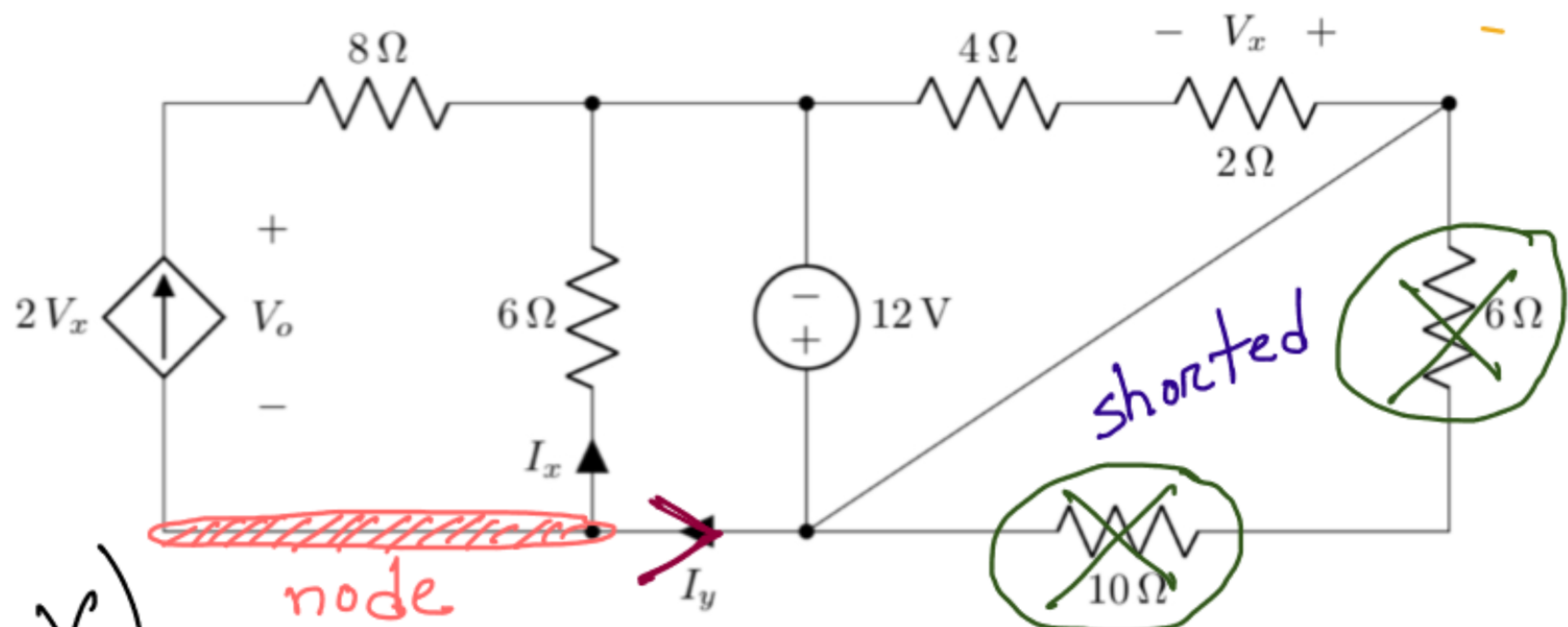


Set A

Question: 01

(a) VDR $\rightarrow$

$$V_x = \frac{2}{2+4} \cdot (12V) \\ = 4 \text{ volts}$$



Now, given $P_{2V_x} = -416W$

$$\Rightarrow -V_o \times (2V_u) = -416W$$

$$\Rightarrow V_o = \frac{16}{2 \times 4}$$

$$\therefore V_o = 52 \text{ volts (ans)}$$

(b) $I_x = \frac{12}{6}$

$$I_x = 2A \text{ (ans)}$$

$$\therefore 2V_u = 2 \times 4 \\ = 8A$$

Now, KCL at node $\rightarrow 0 = I_y + I_u + 2V_u$

$$\therefore I_y = -(2+8)$$

$$\therefore I_y = -10A \text{ (ans)}$$

(incoming = outgoing)

Question: 02

(a) Using Nodal Method

$$V_1 - V_{\text{Ref}} = 20\text{V}$$

$$\therefore V_1 = 20 \text{ volts}$$

$$V_2 - V_3 = 2V_y \text{ (supernode)}$$

$$\Rightarrow V_2 - V_3 = 2[V_3 - V_1]$$

$$[V_y = V_3 - V_1]$$

$$\Rightarrow V_2 - V_3 = 2V_3 - 2V_1$$

$$[V_1 = 20 \text{ volts}]$$

$$\Rightarrow V_2 - V_3 = 2V_3 - 40$$

$$\Rightarrow -V_2 + 3V_3 = 40 \text{ (i)}$$

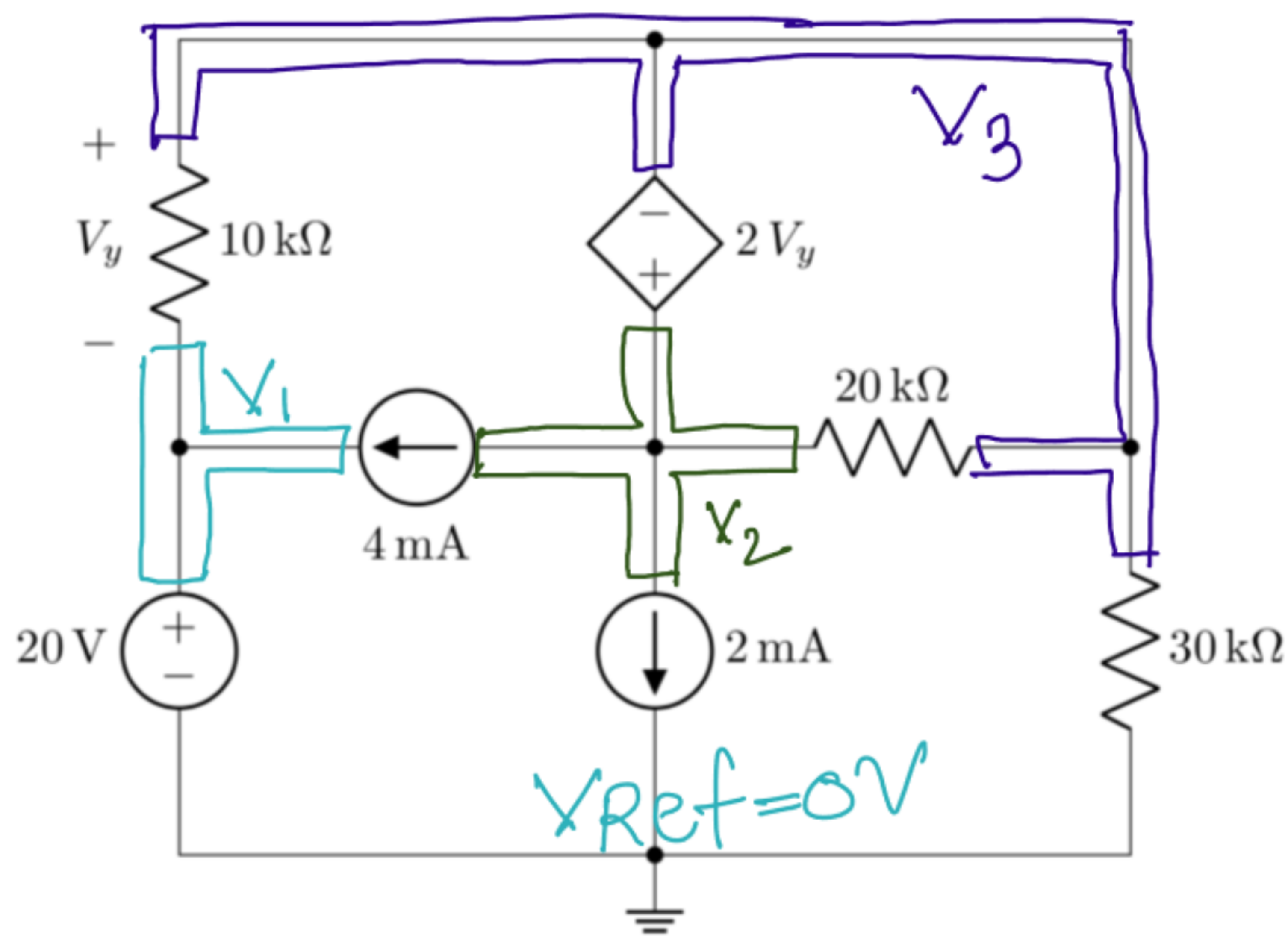
KCL at supernode $\rightarrow$

$$4 + 2 + \frac{V_2 - V_3}{20} + \frac{V_3 - V_2}{20} + \frac{V_3 - 0}{30} + \frac{V_3 - V_1}{10} = 0$$

$$\Rightarrow \frac{V_3 + 3V_3 - 3V_1}{30} = -6$$

$$\Rightarrow 4V_3 - 3V_1 = -180 \quad \xrightarrow{20\text{V}}$$

$$\Rightarrow 4V_3 = -180 + 60 \text{ (ii)}$$

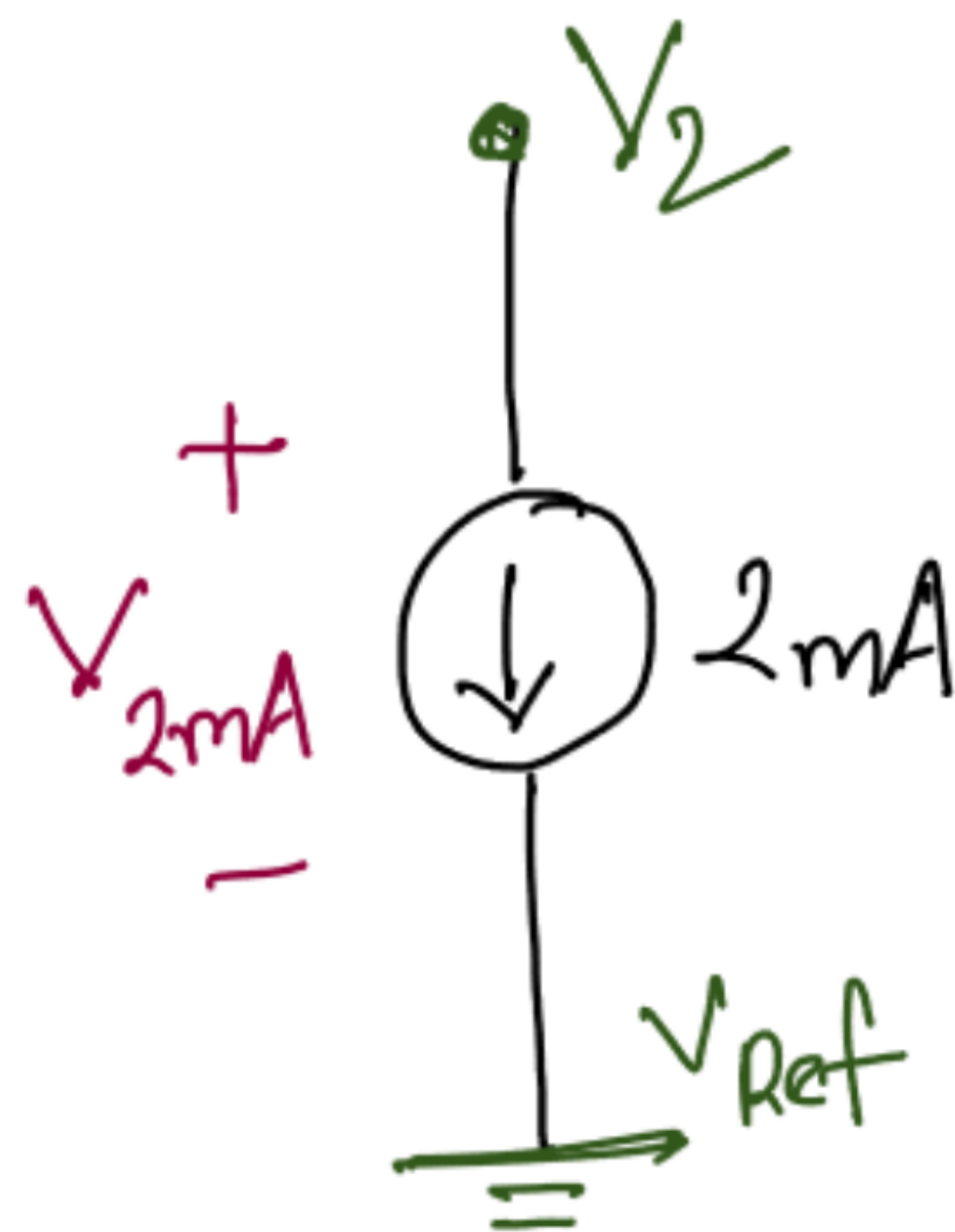


Solving (i) and (ii) we get $\Rightarrow$

$$V_2 = -130 \text{ volts}$$

$$V_3 = -30 \text{ volts}$$

(i) Power of 2mA source



$$V_{2mA} = V_2 - V_{Ref}$$

$$\Rightarrow V_{2mA} = -130 - 0$$

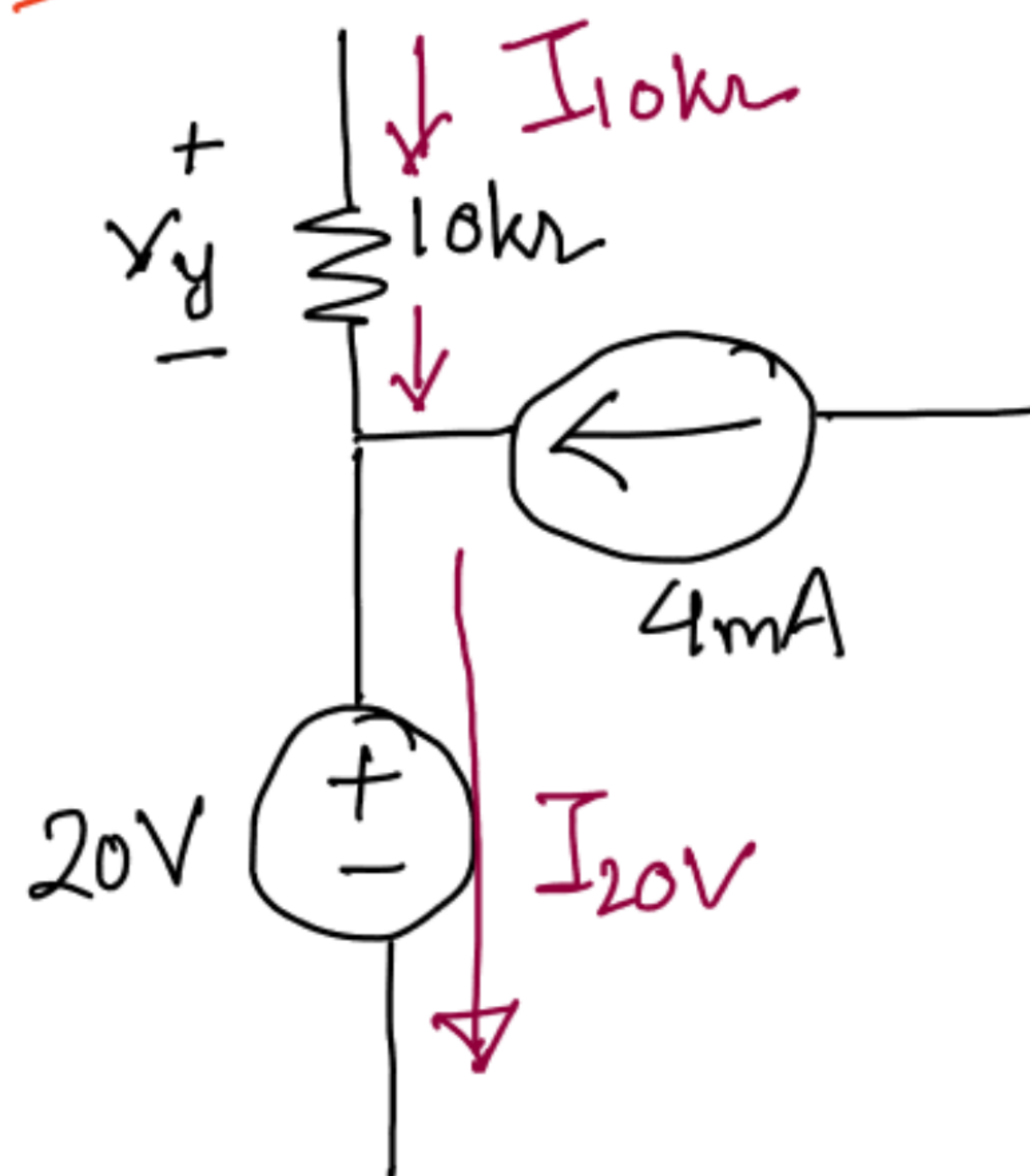
$$\therefore V_{2mA} = -130 \text{ volts}$$

$$P_{2mA} = +(-130V) \cdot (2mA)$$

$$= -260 \text{ mW}$$

(supplying)

(ii) Power of 20V source



$$I_{10k} = \frac{V_y}{10k\Omega} = \frac{V_3 - V_1}{10k\Omega}$$
$$= \frac{-30 - 20}{10k\Omega}$$

$$I_{10k} = -5 \text{ mA}$$

$$I_{20V} = I_{10k} + 4mA$$
$$= -1 \text{ mA}$$

$$\therefore P_{20V} = +(20V) \cdot (-1mA)$$

$$= -20 \text{ mW}$$

(supplying)

(b) Using Mesh Analysis $\rightarrow$

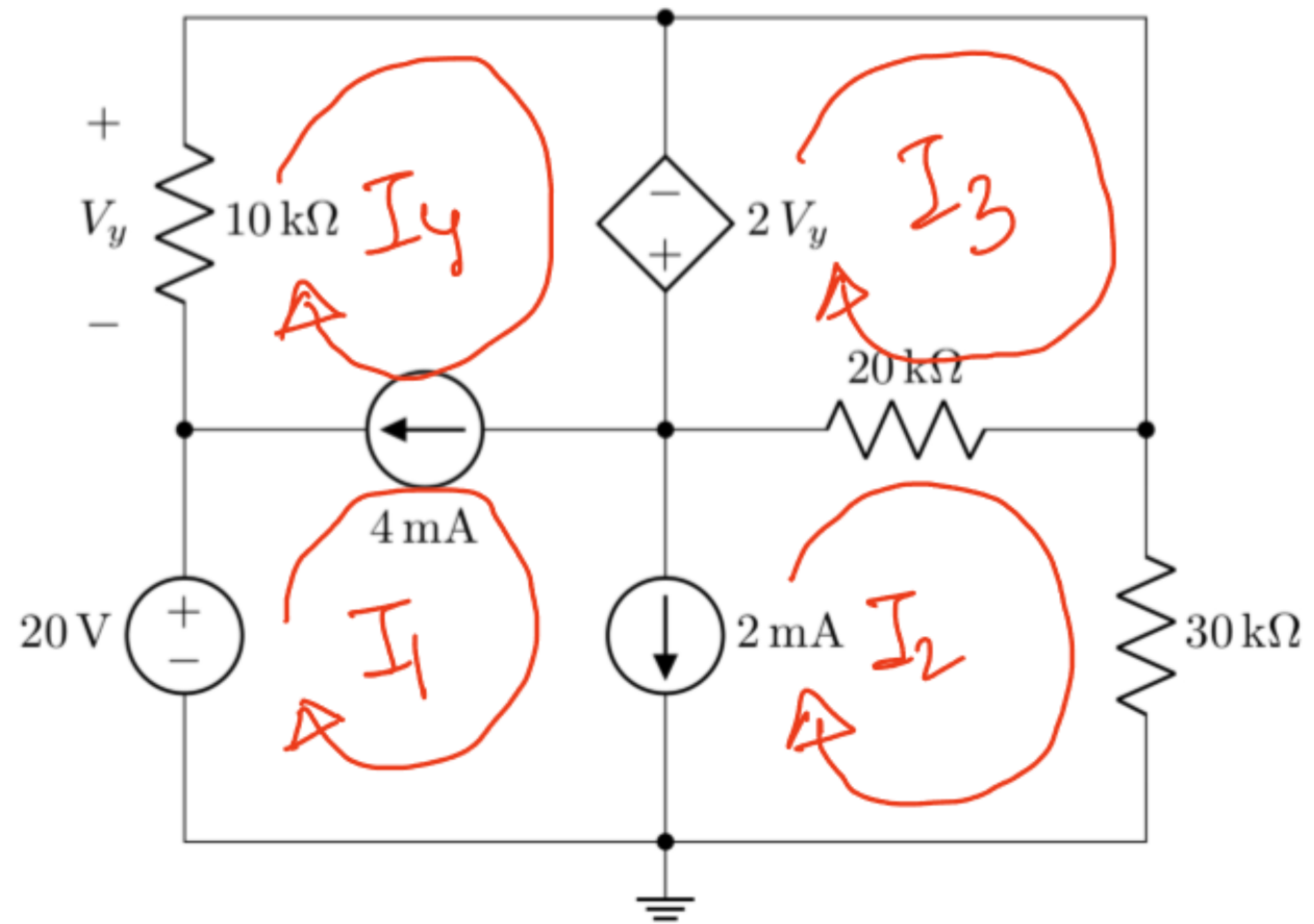
① $I_1 - I_2 = 2$

② $I_4 - I_1 = 4$

③ $30I_2 - 20 + 10I_4 - 2V_y + 20(I_2 - I_3) = 0$

$\rightarrow V_y = -10I_4$

④ $+2V_y + 20(I_3 - I_2) = 0$

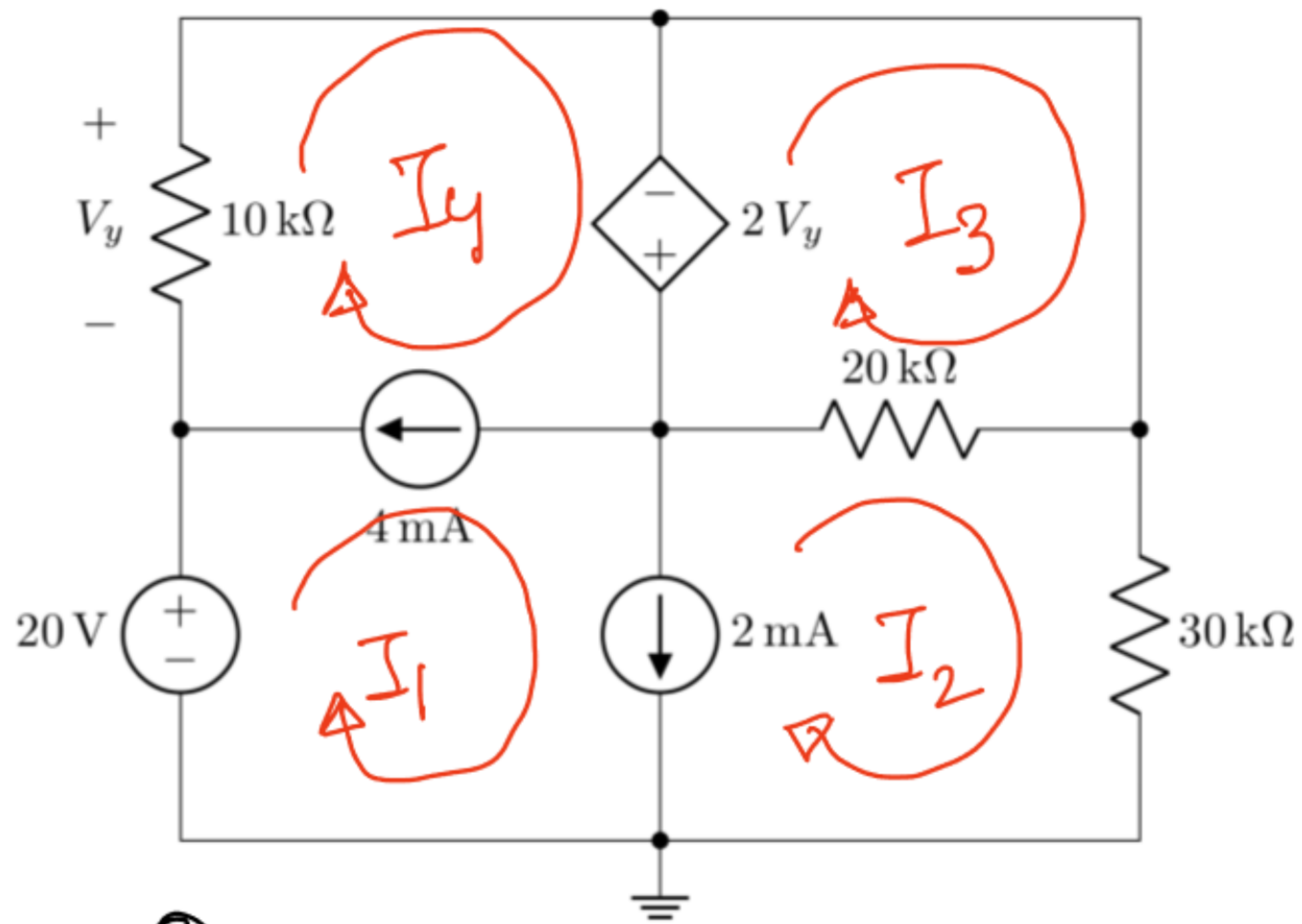


Another approach can be found at the next page of this solution pdf

Question: 02 (another)

Using Mesh Method

KVL at non-supermesh loop $\rightarrow$



$$20(I_3 - I_2) + 2V_y = 0$$

$$\Rightarrow 20I_3 - 20I_2 + 2 \times (-10I_4) = 0$$

$$\Rightarrow -20I_2 + 20I_3 - 20I_4 = 0$$

$$\Rightarrow I_2 - I_3 + I_4 = 0 \dots\dots (i)$$

Now, $I_1 - I_2 = 2$; $I_4 - I_1 = 4$

$\dots\dots (ii)$ $\dots\dots (iii)$

KVL $\rightarrow$

$$-20 + 10I_4 - 2V_y + 20(I_2 - I_3) + 30I_2 = 0$$

-10I₄

$$\Rightarrow 50I_2 - 20I_3 + 30I_4 = 20$$

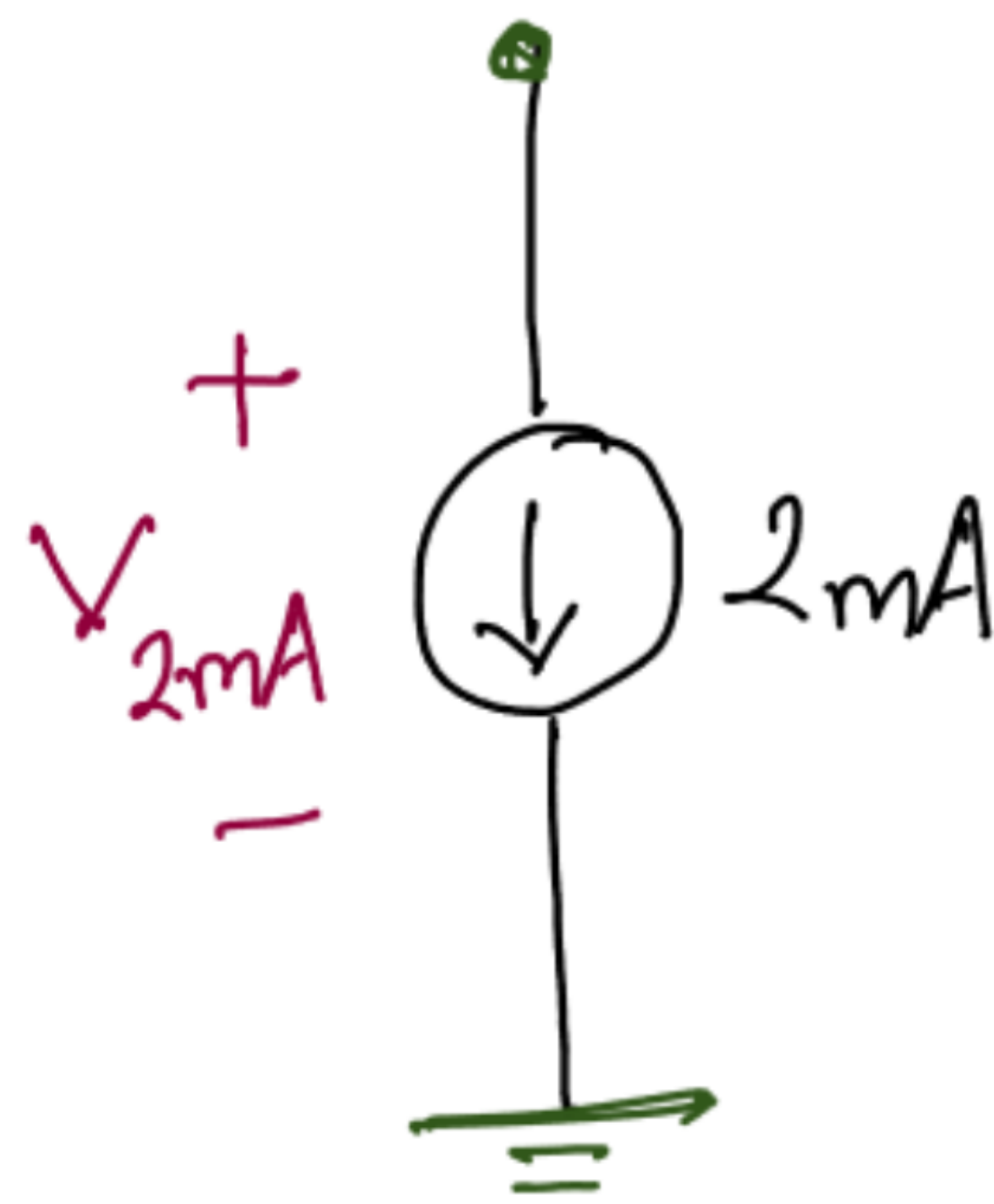
$\dots\dots (iv)$

Solving (i), (ii), (iii) and (iv) $\rightarrow$ we get:

$$I_1 = 1 \text{ mA} ; I_2 = -1 \text{ mA} ; I_3 = 4 \text{ mA}$$

$$I_4 = 5 \text{ mA}$$

(i) Power of 2mA source



KVL at Mesh-2 $\Rightarrow$

$$30I_2 - V_{2\text{mA}} + 20(I_2 - I_3) = 0$$

$\downarrow$ $\downarrow$ $\swarrow$
 -1mA -1mA 4mA

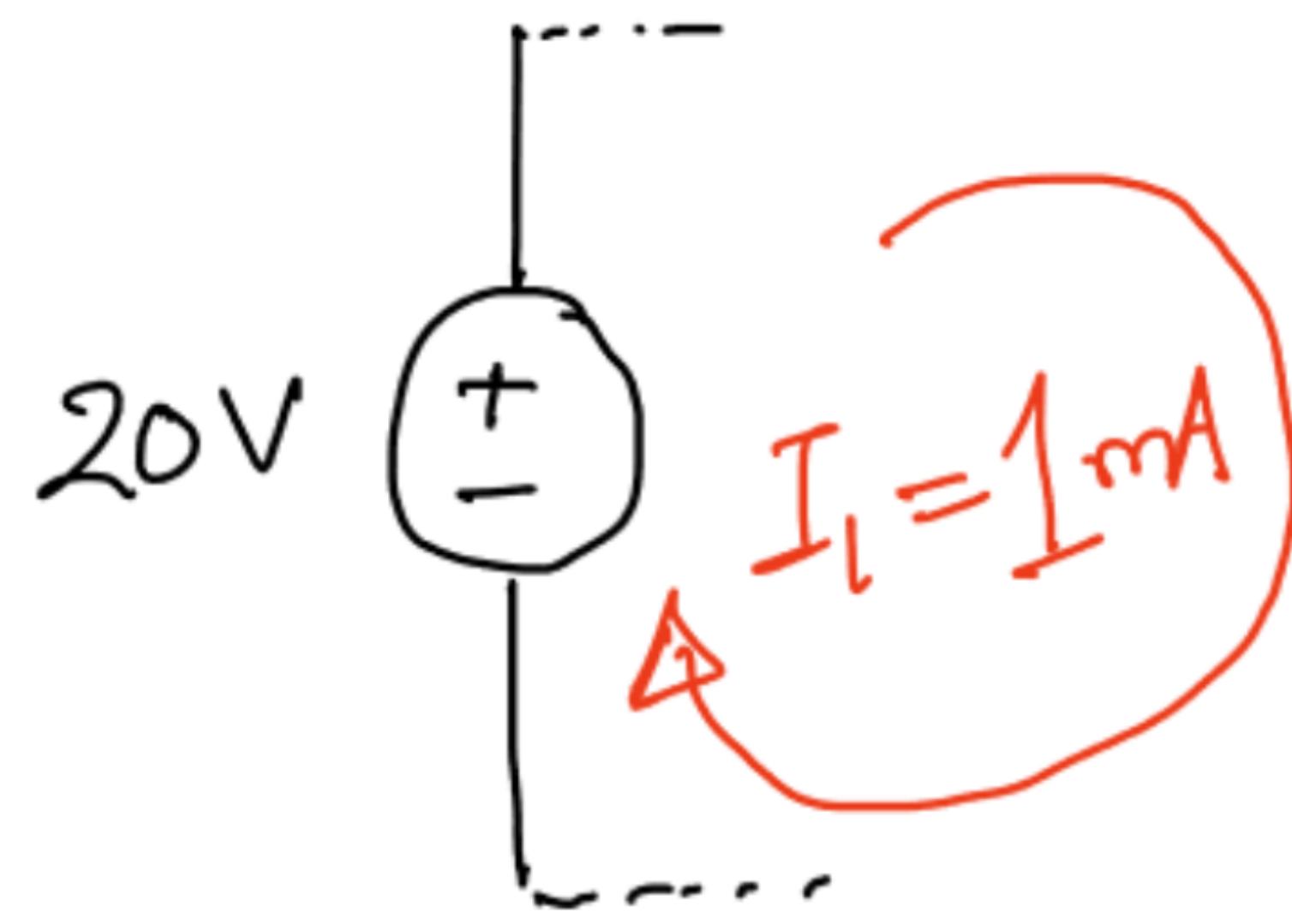
$$V_{2\text{mA}} = -130 \text{ volts}$$

$$P_{2\text{mA}} = +(-130 \text{ V}) \cdot (2 \text{ mA})$$

$$= -260 \text{ mW}$$

(supplying)

(ii) Power of 20V source



$$\therefore P_{20\text{V}} = -(20\text{V}) \cdot (1\text{mA})$$
$$= -20 \text{ mW}$$

(supplying)

Question: 03 →

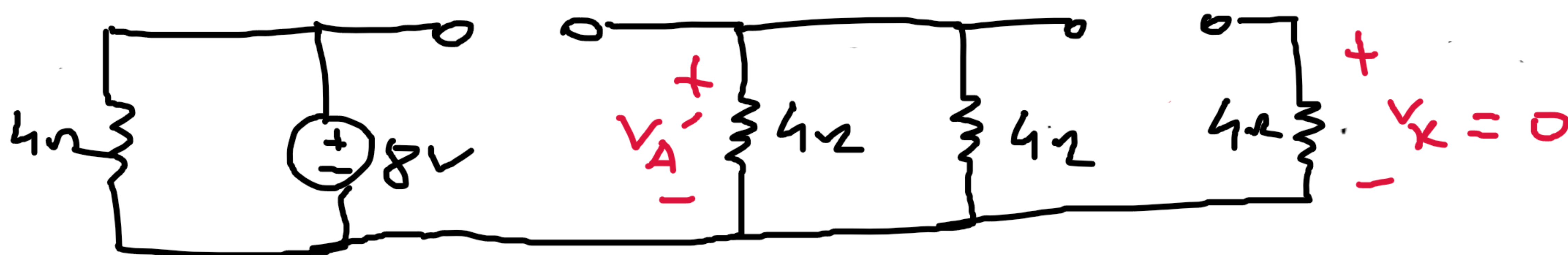
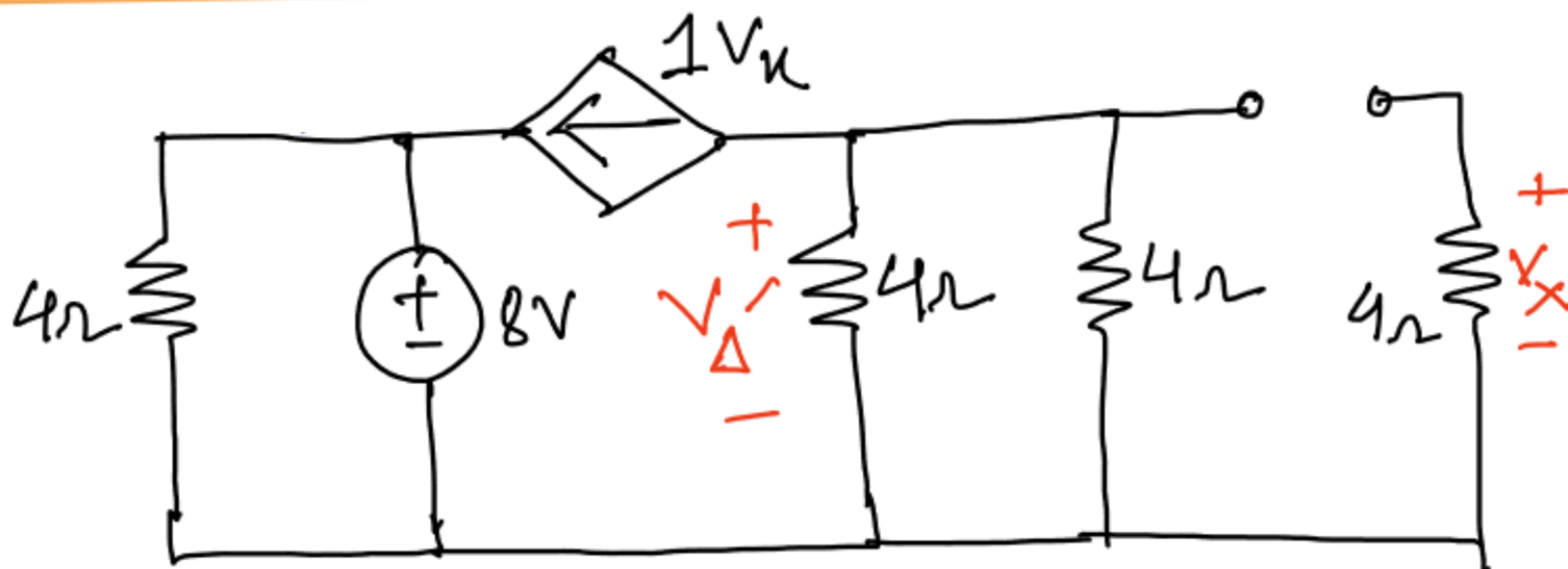
Superposition for 8V source →

$$V_x = 0$$

So, dependent

current source

$$1V_x = 0 \text{ [open]}$$



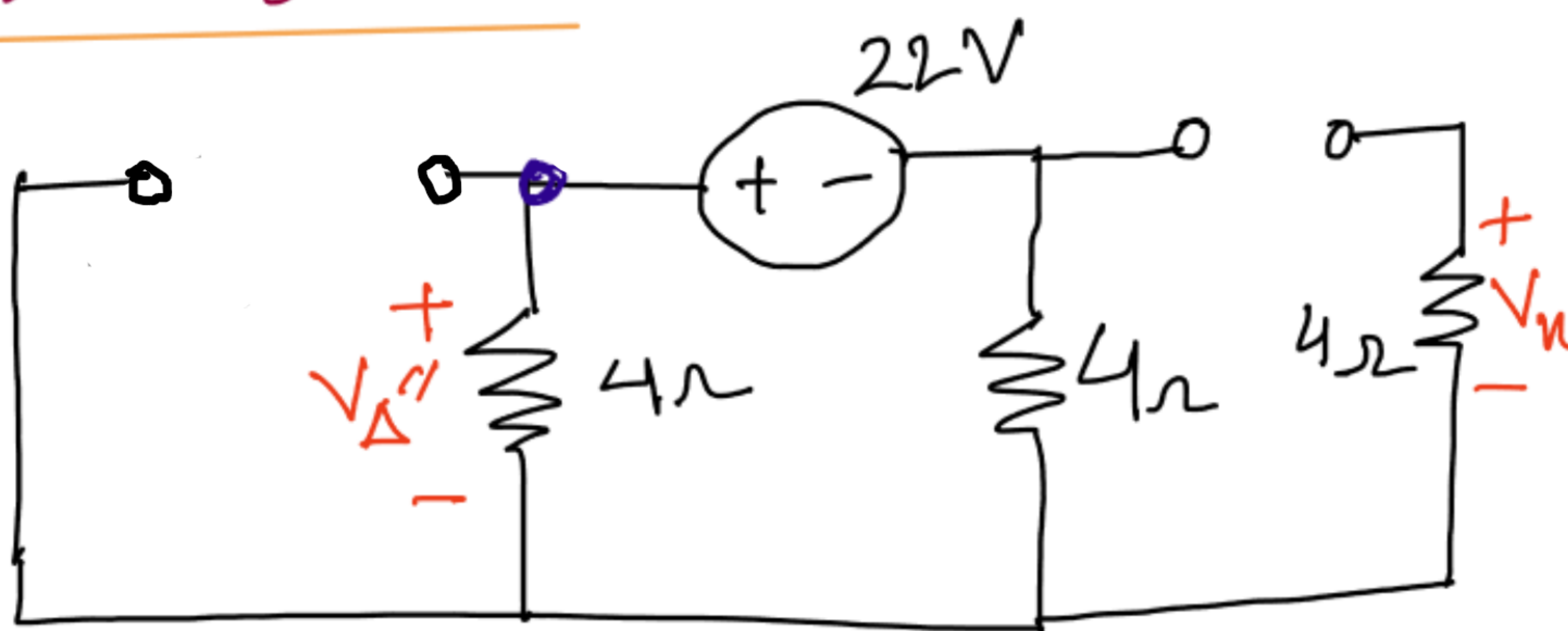
Eventually,

$$V_{\Delta}' = 0V$$

Superposition for 22V source →

Again $V_x = 0$ and

hence $1V_x = 0$



voltage division,

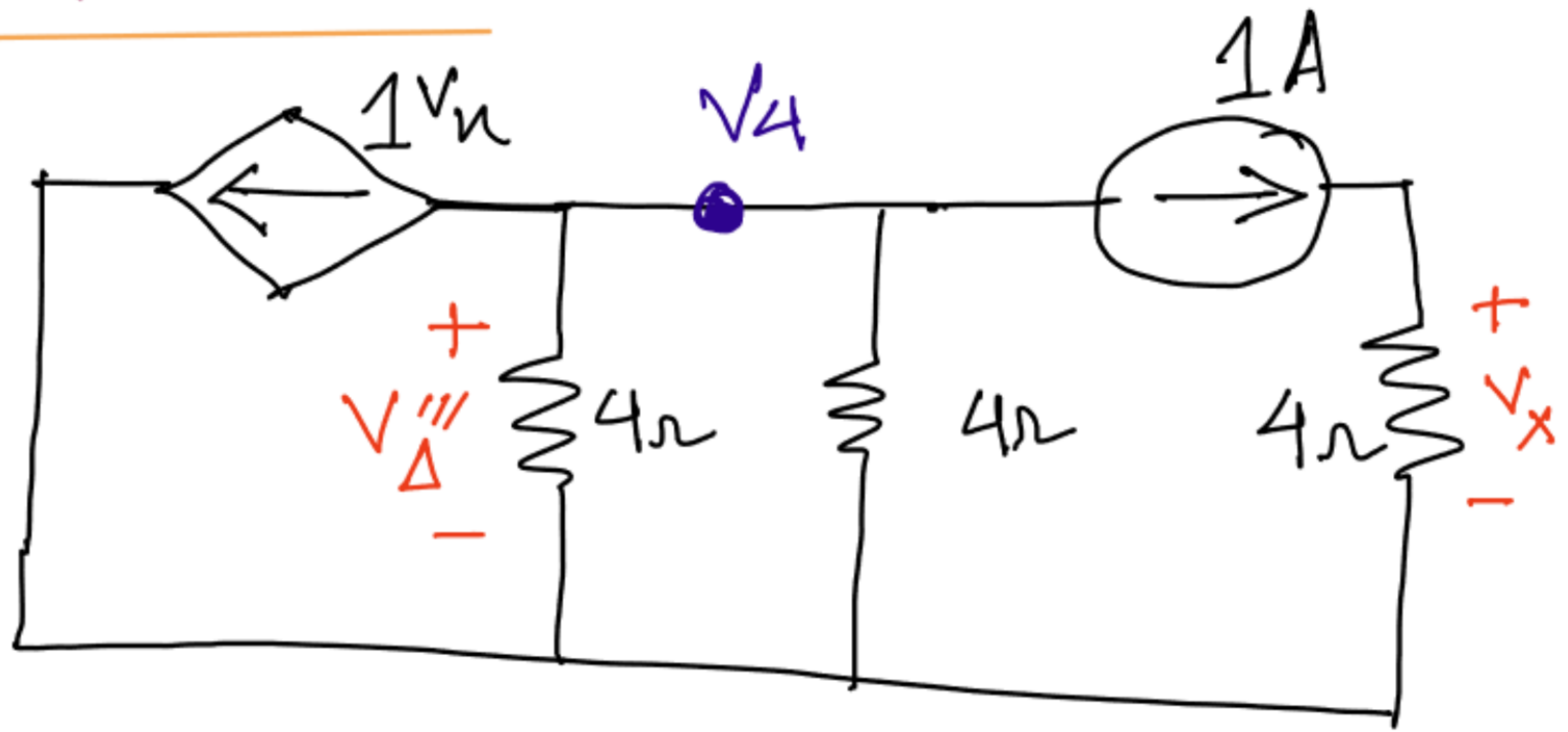
$$V_{\Delta}'' = \frac{4}{4+4} \times 22 \rightarrow$$

$$V_{\Delta}'' = 11 \text{ volts}$$

Superposition for 1A source $\rightarrow$

here, $V_n = (4\Omega)(1A)$

$\therefore V_n = 4 \text{ volts}$



Nodal Analysis $\rightarrow$

$$1V_n + \frac{V_4 - 0}{4} + \frac{V_4 - 0}{4} + 1 = 0$$

$\searrow$ 4 volts

$$\Rightarrow \frac{V_4}{2} = -5$$

$$\Rightarrow V_4 = -10 \text{ volts}$$

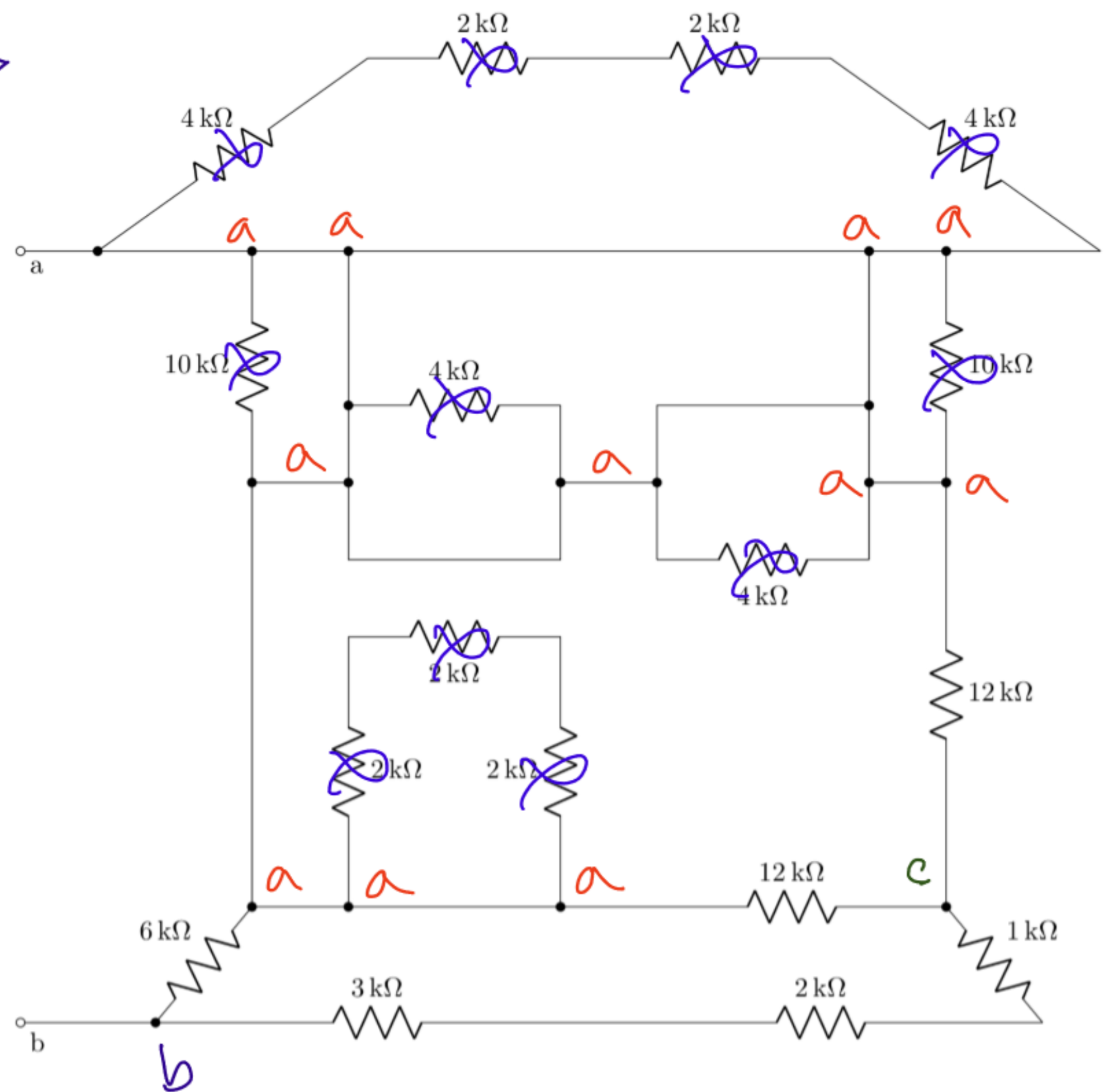
$$V_{\Delta}''' = V_4 - V_{Ref}$$

$$\therefore V_{\Delta}''' = -10 \text{ volts}$$

$$\begin{aligned} \text{total response, } V_{\Delta} &= V_{\Delta}' + V_{\Delta}'' + V_{\Delta}''' \\ &= 0 + 11 - 10 \end{aligned}$$

$$\therefore V_{\Delta} = 1 \text{ volts}$$

Question : 04 →



$$R_{ac} = 12 \parallel 12$$

$$= 6 \text{ k}\Omega$$

$$R_{ab} = (6 + 6) \parallel 6$$

$$R_{ab} = 4 \text{ k}\Omega$$

